→ Find $\int_0^\infty e^{-x^2} \cdot dx$

$x^2 = t$

$2x\,dx = dt$

$\int_0^\infty e^{-t} \cdot \frac{dt}{2x}$

$= \frac{1}{2} \int_0^\infty x^{-1} \cdot e^{-t} \cdot dt$

$= \frac{1}{2} \int_0^\infty x^{0-1} \cdot e^{-t} \cdot dt$

$= \frac{1}{2} \Gamma(0) = 1/2$

→ Find $\int_0^\infty e^{-x^2/2} dx$

$\frac{x^2}{2} = t$

$\frac{2x\,dx}{2} = dt$

$x\,dx = dt$

$\int_0^\infty e^{-t} \cdot \frac{dt}{x}$

$2\sqrt{x}$

$= \int_0^\infty x^{-1} \cdot e^{-t} dt$

$= \Gamma(0) = 1$

→ Find $\int_0^\infty e^{-x^2/2} dx$

$\frac{x^2}{2} = t \Rightarrow \frac{2x\,dx}{2} = dt$

$\Rightarrow dx = \frac{dt}{\sqrt{2t}}$

$\int_0^\infty e^{-t} \cdot \frac{dt}{\sqrt{2t}} = \frac{1}{\sqrt{2}} \int_0^\infty t^{\frac{1}{2}-1} \cdot e^{-t} \cdot dt$

$= \frac{1}{\sqrt{2}} \Gamma\left(\frac{1}{2}\right)$

$= \frac{\sqrt{\pi}}{\sqrt{2}} = \sqrt{\frac{\pi}{2}}$

11/2/20.

Transform Methods

Moment Generating Function

Probability Generating Function

Charactorstic function.

$M_X(t) = E(e^{tX}) = \begin{cases} \sum_i p(x_i) e^{tx_i}, & \text{if } X \text{ is discrete} \\ \int f(x) e^{tx} dx, & \text{if } X \text{ is continuous} \end{cases}$

↓ ~~Moment~~ Moment Generating Function (mgf)

(*)

$E[e^{tX}] = E\left[1 + tX + \frac{t^2}{2!} X^2 + \frac{t^3 X^3}{3!} + \cdots\right]$

$= 1 + t \cdot E(X) + \frac{t^2}{2!} E[X^2] + \cdots$

→ Ex:- $X \sim Bin(n, \theta)$ (Binomial)

$P_X(i) = {}^nC_i\, \theta^i (1-\theta)^{n-i}$

$i = 0, 1, \ldots n$

mgf of X

$M_X(t) = E(e^{tX})$

$= \sum_{i=0}^{n} p_i e^{ti}$

$$\sum_{i=0}^{n} {}^nC_i \theta^i (1-\theta)^{n-i} \cdot e^{ti}$$

$$= \sum_{i=0}^{n} {}^nC_i (\theta e^t)^i (1-\theta)^{n-i}$$

$$\Rightarrow \boxed{M_X(t) = (\theta e^t + (1-\theta))^n}$$ (Binomial)

$\rightarrow M_X(t) = E[e^{tX}]$ (First Moment)

$$\left(\frac{d}{dt} M_X(t)\right)_{t=0} = \frac{d}{dt} E[e^{tX}]$$

$$= \left(E[X e^{tX}]\right)_{t=0}$$

$$= E[X]$$

$$\rightarrow \frac{d}{dt}(M_X(t)) = E(X e^{tX})$$

$$\frac{d}{dt}\left(\frac{d}{dt}(M_X(t))\right) = E(X^2 e^{tX})$$

$$\left(\frac{d^2}{dt^2}(M_X(t))\right)_{t=0} = E[X^2]$$

(Second Moment)

Now for previous Ex

$$M_X(t) = (\theta e^t + (1-\theta))^n$$

$$\frac{d}{dt}(M_X(t)) = \frac{d}{dt}(\theta e^t + (1-\theta))^n$$

$$\frac{d}{dt}(M_X(t)) = n(\theta e^t + (1-\theta))^{n-1} \cdot \theta e^t$$

$$\left(\frac{d}{dt}(M_X(t))\right)_{t=0} = n\theta$$

$$\Rightarrow \boxed{E(X) = n\theta}$$

Now

$$\frac{d^2}{dt^2}(M_X(t)) = \frac{d}{dt}\left(n(\theta e^t + (1-\theta))^{n-1} \cdot \theta e^t\right)$$

$$= n\theta\left[e^t(\theta e^t + (1-\theta))^{n-1} + e^t(n-1)(\theta e^t + (1-\theta))^{n-2} \cdot \theta e^t\right]$$

$$\left(\frac{d^2}{dt^2}(M_X(t))\right)_{t=0} = n\theta[1 + (n-1)\cdot\theta]$$

$$= n\theta + n(n-1)\theta^2$$

$$\Rightarrow \boxed{E(X^2) = n\theta + n(n-1)\theta^2}$$

$$\sigma_X^2 = E(X^2) - (E(X))^2$$

$$= n\theta + n(n-1)\theta^2 - n^2\theta^2$$

$$= n\theta + n^2\theta^2 - n\theta^2 - n^2\theta^2$$

$$= n\theta - n\theta^2$$

$$\boxed{\sigma_X^2 = n(\theta)(1-\theta)}$$

$\rightarrow$ Ex: $X \sim P(\lambda)$ (Poissons)

$$P_i = \frac{e^{-\lambda} \cdot \lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

~~mgf of~~ x

$$M_X(t) = E[e^{tX}]$$

$$= \sum_{i=0}^{\infty} P_i \cdot e^{ti}$$

$$= \sum_{i=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^i}{i!} \cdot e^{ti} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i \cdot e^{ti}}{i!}$$

$$= e^{-\lambda} \cdot \left[1 + \frac{\lambda e^t}{1!} + \frac{\lambda^2 e^{2t}}{2!} + \cdots \right]$$

$$= e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

$$\boxed{M_x(t) = e^{\lambda(e^t-1)}}$$ of Poisson

$$\frac{d}{dt}(M_x(t)) = e^{\lambda(e^t-1)} \cdot (\lambda(e^t))$$

$$\left(\frac{d}{dt}(M_x(t))\right)_{t=0} = \lambda$$

$$\Rightarrow \boxed{E[x] = \lambda}$$

$$\frac{d^2}{dt^2}(M_x(t)) = \frac{d}{dt}\left(\lambda \cdot e^{\lambda e^t - \lambda + t}\right)$$

$$= \lambda \cdot e^{\lambda e^t - \lambda + t}(\lambda e^t + 1)$$

$$\left(\frac{d^2}{dt^2}(M_x(t))\right)_{t=0} = \lambda \cdot (\lambda + 1)$$

$$\Rightarrow \boxed{E(x^2) = \lambda(\lambda+1)}$$

$$\sigma_x^2 = E(x^2) - (E(x))^2$$

$$= \lambda(\lambda+1) - \lambda^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$\boxed{\sigma_x^2 = \lambda}$$

13/2/20

MGF

x is a r.v

$$M_x(\theta) = E[e^{\theta x}]$$

$$= \begin{cases} \sum_{i=1}^{n} e^{\theta x_i} p(x_i) & \text{, if discrete} \\ \int e^{\theta x} f(x)\, dx & \text{, if continuous} \end{cases}$$

$\rightarrow$ $x \sim Bin(n, \frac{1}{4})$

Find $E(e^{x+7})$

$$= E(e^x \cdot e^7)$$

$$\cancel{= e^7 \cdot n\theta = \frac{n e^7}{4}}$$

$$= e^7 \cdot E(e^x)$$

$$= e^7 \cdot \sum p_i e^i$$

$$= e^7 \sum {}^nC_i \left(\frac{1}{4}\right)^i \left(\frac{3}{4}\right)^{n-i} e^i$$

$$= e^7 \sum {}^nC_i \left(\frac{e}{4}\right)^i \left(\frac{3}{4}\right)^{n-i}$$

$$= e^7 \left(\frac{e}{4} + \frac{3}{4}\right)^n$$

$$E(e^{x+7}) = e^7 \left(\frac{3+e}{4}\right)^n$$

$\rightarrow$ x is a bernoulli rv

$p(x=1) = \theta$

$p(x=0) = 1-\theta$

Find mgf & $E(X)$?

~~mgf = $E[e^{tX}]$~~

~~$M_X(t) = E[e^{tX}]$~~

X:	0	1
P_i:	$1-\theta$	θ

$$E[e^{tX}] = (1-\theta)e^{t\cdot 0} + e^{t\cdot 1}\cdot\theta$$

$$= (1-\theta) + \theta e^{t}$$

$$M_X(t) = (1-\theta) + \theta e^{t}$$

~~$E(X) = \frac{d}{dt}(M_X(t))$~~

$$E(X) = \left(\frac{d}{dt}(M_X(t))\right)_{t=0}$$

$$= (\theta e^{t})_{t=0}$$

$$E[X] = \theta$$

(or) $E(X) = n\theta$
$= 1\cdot\theta$

$E(X) = \theta$

$\boxed{\text{* Bernoulli variate is binomial variate with } n = 1}$

→ A discrete r.v X has mgf

$$M_X(t) = e^{2(e^{t}-1)}$$

Find $P(X=1)$?

This is mgf of poission function where $\lambda = 2$

$$\Rightarrow P(X=r) = \frac{e^{-\lambda}\cdot\lambda^{r}}{r!}$$

$$P(X=1) = \frac{e^{-2}\cdot(2)^{1}}{1!} = 2e^{-2}$$

→ $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty$$

$M_X(t) = ?$

$$= E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} e^{tx}\cdot f(x)\cdot dx$$

$$= \int_{-\infty}^{\infty} e^{tx}\,\frac{e^{-(x-\mu)^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma}\,dx$$

$$\boxed{M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}}$$ How?

Let $\frac{x-\mu}{\sigma} = y$

$dx = \sigma\, dy$

$$\Rightarrow M_X(t) = \int_{-\infty}^{\infty} \frac{e^{t(\sigma y+\mu)}\cdot e^{-y^2/2}}{\sqrt{2\pi}\,\cancel{\sigma}}\cdot\cancel{\sigma}\,dy$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\, e^{t(\mu+\sigma y)}\cdot e^{-y^2/2}\,dy$$

$$= \frac{1}{\sqrt{2\pi}}\, e^{\mu t}\int_{-\infty}^{\infty} e^{\sigma y t}\cdot e^{-y^2/2}\,dy$$

$$= e^{\mu t}\cdot\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(y^2-2\sigma y t+\sigma^2 t^2)}\cdot e^{\frac{1}{2}\sigma^2 t^2}\cdot dy$$

$$= e^{\mu t}\cdot\frac{1}{\sqrt{2\pi}}\, e^{\frac{1}{2}\sigma^2 t^2}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(y-\sigma t)^2}\cdot dy$$

$$= \frac{1}{\sqrt{2\pi}}\, e^{\mu t + \frac{\sigma^2 t^2}{2}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(y-\sigma t)^2}\cdot dy$$

Put $y - \sigma t = z$

$dy = dz$

$$M_x(t) = \frac{1}{\sqrt{2\pi}} e^{\mu t + \frac{\sigma^2 t^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z)^2} \cdot dz$$

After simplifying

$$M_x(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$Z \sim N(0,1)$

$$\boxed{M_x(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}} \quad \text{where } X \sim N(\mu, \sigma^2)}$$

$E(x) = ?$

$V(x) = ?$

$$\frac{d}{dt}(M_x(t))_{t=0} = \left(e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot \left(\mu + \frac{\sigma^2 \cdot 2t}{2}\right)\right)_{t=0}$$

$$= 1 \cdot (\mu)$$

$$\boxed{E(x) = \mu}$$

$$\frac{d^2}{dt^2}(M_x(t))_{t=0} = \left(\left(\mu + \sigma^2 t\right)^2 \left(e^{\mu t + \frac{\sigma^2 t^2}{2}}\right) + e^{\mu t + \frac{\sigma^2 t^2}{2}} (\sigma^2)\right)_{t=0}$$

$$\boxed{E(x^2) = \mu^2 + \sigma^2}$$

$$\sigma_x^2 = E(x^2) - [E(x)]^2$$

$$= \mu^2 + \sigma^2 - \mu^2$$

$$\boxed{\sigma_x^2 = \sigma^2}$$

$\rightarrow Z = \frac{X - \mu}{\sigma}$ Standard Variate

$$E[Z] = \frac{1}{\sigma}(E(x) - \mu)$$

$$= \frac{\mu - \mu}{\sigma} = 0$$

$$\boxed{E(Z) = 0}$$

$$V(Z) = E(Z^2) - (E(Z))^2$$

$$= E\left(\left(\frac{X-\mu}{\sigma}\right)^2\right) - 0$$

$$= \frac{1}{\sigma^2} E(x^2 + \mu^2 - 2\mu x)$$

$$= \frac{1}{\sigma^2}\left[E(x^2) + \mu^2 - 2\mu^2\right]$$

$=$

$X \sim N(\mu, \sigma^2)$

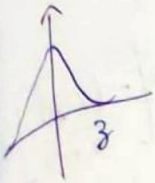
$Z = \frac{X - \mu}{\sigma}$

$E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = 0$

$V(Z) = E\left(\left(\frac{X-\mu}{\sigma}\right)^2\right) = 1$

$Z \sim N(0,1)$

NOTE:
$Z \sim N(0,1)$



$$F_Z(z) = P(Z \le z)$$
$$= \Phi(z)$$

Central Moments

about mean μ

$$\mu_r = E(X-\mu), \quad r = 1, 2, \ldots$$

$X \sim N(\mu, \sigma^2)$

$E((X-\mu)^3)$ Third moment

Movement about origin

$$\mu_r' = E[X^r]$$

18/2/20

→ X is a cont r.v with pdf

$$f(x) = \begin{cases} ax & 0 \le x < 1 \\ a & 1 \le x < 2 \\ -ax+3a & 2 \le x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

Find a and compute $P(X \le 1.5)$

$\int_0^3 f(x)\,dx = 1$ ∵ pdf

$\Rightarrow \int_0^1 ax\,dx + \int_1^2 a \cdot dx + \int_2^3 -ax+3a = 1$

$\Rightarrow \frac{a}{2} + a - \frac{a}{2}[5] + 3a(1) = 1$

$\Rightarrow$ ~~$\frac{a}{2} + 15 = 1$~~ $\Rightarrow$ ~~$15 = 2a$~~

$\Rightarrow$ ~~$13 = 2a$~~

$\Rightarrow$ ~~$a = 13/2$~~

$\Rightarrow \frac{3a - 5a + 6a}{2} = 1$

$\Rightarrow 4a/2 = 1$

$a = 1/2$

$P(X \le 1.5)$

$= \int_0^{1.5} f(x) \cdot dx$

$= \int_0^1 f(x)\,dx + \int_1^{1.5} f(x)\,dx$

$= \int_0^1 ax\,dx + \int_1^{1.5} a\,dx$

$= \frac{a}{2} + a(0.5)$

$= \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$

$P(X \le 1.5) = 1/2$

→ Let G represent Geometric mean

$\log G = \int \log x \cdot f(x)\,dx$

Given $f(x) = 6(2-x)(x-1) \quad 1 \le x \le 2$

$\Rightarrow \log G = \int \log x \cdot 6(2-x)(x-1)\,dx$

$= 6 \int_1^2 \log x\,(2-x)(x-1) \cdot dx$

$= 6 \int \log x\,(2x - 2 - x^2 + x) \cdot dx$

$= 6 \int 3x \log x - x^2 \log x - 2 \log x \cdot dx$

$= 6\left[3\left(\log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2}\,dx \right) - \left(\log x \cdot \frac{x^3}{3} - \int \frac{1}{x} \cdot \frac{x^3}{3} \right) - 2(x \log x - x) \right)$

$= 6\left[3\left(\frac{x^2}{2} \log x - \frac{1}{2} \cdot \frac{x^2}{2} \right) - \left(\frac{x^3 \log x}{3} - \frac{1}{3} \cdot \frac{x^3}{3} \right) - 2(x \log x - x) \right)_1^2$

$$= 6\left(3\left[2\log 2 - 1\right] - \left[\frac{8\log 2}{3} - \frac{8}{9}\right] - 2\left[2\log 2 - 2\right] + \left[3\left(0 - \frac{1}{4}\right) - \left(0 - \frac{1}{9}\right) - 2\left(0 - 1\right)\right]\right)$$

$$= 6\left(6\log 2 - 3 - \frac{8\log 2}{3} + \frac{8}{9} - 4\log 2 + 4 - \left(-\frac{3}{4} + \frac{1}{9} + 2\right)\right)$$

$$= 6\left(6\log 2 - 3 - \frac{8\log 2}{3} + \frac{8}{9} - 4\log 2 + 4 + \frac{3}{4} - \frac{1}{9} - 2\right)$$

$$= 6\left(2\log 2 - \frac{8\log 2}{3} + 1 + \frac{3}{4} + \frac{7}{9} - 2\right)$$

$$= 6\left(\frac{-2\log 2}{3} + \frac{27+28}{36} - 1\right)$$

$$= 6\left(\frac{-2\log 2}{3} + \frac{19}{36}\right)$$

$$\boxed{\ln G = \frac{19}{6} - 4\log 2}$$

$$\Rightarrow \boxed{\ln G + 4\ln 2 = \frac{19}{6}}$$

$$\boxed{\ln(16G) = 19/6}$$

→ $\boxed{\text{PDF}}$

1.) Uniform — pdf

$X \sim U(a,b)$ $\quad f(x) = \frac{1}{b-a}$ $\quad a \le x \le b$

2.) Exponential

$X \sim E(\lambda)$ $\quad f(x) = \lambda e^{-\lambda x}$ $\quad x \ge 0,\ \lambda > 0$

3.) $X \sim$ Gamma $\quad \frac{\lambda \cdot e^{-\lambda x} (\lambda x)^{\alpha - 1}}{\Gamma(\alpha)}$

4.) Normal

$X \sim N(\mu, \sigma^2)$ $\quad \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$

$-\infty < x < \infty$
$-\infty < \mu < \infty$
$\sigma > 0$

5.) Weibull (Failure distribution)

$f(t) = \lambda \alpha x^{\alpha - 1} \cdot e^{-\lambda x^{\alpha}}$

$F(t) = 1 - e^{-\lambda t^{\alpha}}$

Limiting Distribution

Binomial, $M_X(t) = (pe^t + q)^n$

Poisson (Law of Rare Events), $M_X(t) = e^{\lambda(e^t - 1)}$

As $n \to \infty$, $p \to 0$ such that $np = \lambda$ then $M_X(t)$ of Binomial goes to $M_X(t)$ of poisson.

→ P.T Mgf of ~~$M_X(t)$~~ Binomial reaches to Mgf of Poission.

$$M_X(t) = (pe^t + q)^n$$

$$= (pe^t + 1 - p)^n$$

$$M_X(t) = (1 + p(e^t - 1))^n$$

$\lambda = np$

$\Rightarrow M_{X^*}(t) = \left(1 + p(e^t - 1)\right)^{\lambda/p}$

$\cancel{\lim_{p\to 0}} \cancel{\left(1 + p(e^t-1)\right)^{\lambda/p}}$

$$\lim_{p\to 0} M_X(t) = \lim_{p\to 0} \left(1 + p(e^t - 1)\right)^{\frac{\lambda}{p}}$$

$n \to \infty$ such that $np = \lambda$

$$= e^{\frac{\lambda}{p} \cdot p(e^t - 1)}$$

$$= e^{\lambda(e^t - 1)}$$

$= M_X(t)$ of Poisson

Standard Binomial

$$Y = \frac{X - np}{\sqrt{npq}}$$

$$\lim_{n\to\infty} M_Y(t) = e^{t^2/2}$$

Do it....

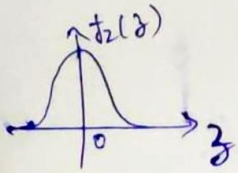
$$M_Y(t) = E\left[e^{tY}\right]$$

$= \mathcal{E}\mu_p$

$\rightarrow$ $Z \sim N(0,1)$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$-\infty < z < \infty$



$$F_Z(z) = \Phi(z) = \int_{-\infty}^{z} f_Z(t)\,dt$$

Gaussian Distribution

Bell-shaped distribution.

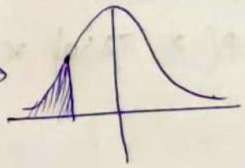
$X \sim N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\Big($$

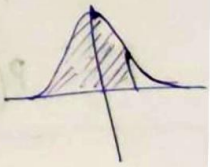
$$Z = \frac{X - \mu}{\sigma}$$

$$\boxed{X = \mu + \sigma Z}$$

$$\rightarrow F_Z(z) = \int_{-\infty}^{z} f_Z(t)\,dt$$



$$\boxed{F_Z(-z) = 1 - F_Z(z)}$$



$$F_Z(-z) = \int_{-\infty}^{-z} f_Z(t)\,dt$$

$$= \int_{z}^{\infty} f_Z(-t)\,dt = \int_{z}^{\infty} f_Z(t)\,d$$

$$= \int_{-\infty}^{\infty} f_Z(t)\,dt - \int_{-\infty}^{z} f_Z(t)\,dt$$

$$F_Z(-z) = 1 - F_Z(z)$$

$\Rightarrow$

$\rightarrow P(a \le Z \le b) = F_Z(b) - F_Z(a)$

Ex:- $X \sim N(200, 256)$

$P(X > 240) = ?$

$$P(X > 240) = 1 - P(X \le 240)$$

$$= 1 - P\left(\frac{X-200}{16} \le \frac{240-200}{16}\right)$$

$$= 1 - P(Z \le 2.5)$$

$\left(\because \frac{X-\mu}{\sigma} = Z\right)$

$= 1 - 0.9938$ (From table we get value of $P(Z \le 2.5)$)

Table will be given in exam.

$P(X > 240) = 0.0062$

$\rightarrow X \sim N(200, 256)$

$\Rightarrow P(X > 240 \mid X > 210)$ Express in cdf of F_Z

$$P(X > 240 \mid X > 210)$$

$$= \frac{P(X > 240 \cap X > 210)}{P(X > 210)}$$

$$= \frac{P(X > 240)}{P(X > 210)}$$

$$= \frac{1 - P(X \le 240)}{1 - P(X \le 210)}$$

$$= \frac{1 - \overset{0.9938}{\cancel{0.0062}}}{1 - P\left(\frac{X-200}{16} \le \frac{210-200}{16}\right)} \rightarrow \text{From previous slide}$$

$$= \frac{\cancel{1 - 0.0062}\ 0.0062}{1 - P(Z \le 0.625)}$$

$$= \frac{\cancel{0.9938}\ 0.0062}{1 - 0.73401}$$

$$= \frac{\cancel{0.9938}\ 0.0062}{0.26599}$$

$P(X > 240 \mid X > 210) = 0.023$

$\rightarrow$ Find

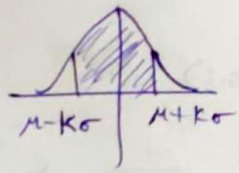
$$P(\mu - k\sigma \le X \le \mu + k\sigma)$$

$$= \frac{2}{\sqrt{\pi}} \int_0^{k/\sqrt{2}} e^{-y^2}\,dy$$

$$= \operatorname{erf}\left(\frac{k}{\sqrt{2}}\right)$$

where

$$\boxed{\operatorname{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^{u} e^{-y^2}\,dy}$$



$P(\mu - k\sigma \le X \le \mu + k\sigma)$

$= F_Z(\mu + k\sigma) - F_Z(\mu - k\sigma)$

$= F_Z(k) - F_Z(-k)$

$= 2F_Z(k) - 1 \quad (\because F_Z(-z) = 1 - F_Z(z))$

$= \frac{2}{\sqrt{2\pi}} \int_{-\infty}^{k} e^{-t^2/2} \cdot dt - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} \cdot dt$

$= \frac{2}{\sqrt{2\pi}} \int_{-\infty}^{k} e^{-t^2/2} dt - \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-t^2/2} \cdot dt$

$= \frac{2}{\sqrt{2\pi}} \int_{0}^{k} e^{-t^2/2} \cdot dt$

$y = \frac{t}{\sqrt{2}}$

$= \frac{2}{\sqrt{\pi}} \int_{0}^{k/\sqrt{2}} e^{-y^2} \cdot dy$

$= \mathrm{erf}\left(\frac{k}{\sqrt{2}}\right)$

→ $P(\mu - 3\sigma \le X \le \mu + 3\sigma)$

$\simeq 0.997$

→ Truncated Normal Density (Non-negative r.v.s)

$$f(x) = \begin{cases} \frac{1}{\alpha\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], & x \ge 0 \\ 0, & x < 0 \end{cases}$$

where

$$\alpha = \int_0^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(\tau-\mu)^2}{2\sigma^2}\right] dt$$

→ NORMAL $\quad X \sim N(\mu, \sigma^2)$

(Additive Mechanism)

→ LOG NORMAL $\quad Y = e^X$

(Multiplicative Mechanism)

$$\rightarrow \phi_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{y} \exp\left[\frac{-(\ln y - \mu)^2}{2\sigma^2}\right]$$

$y > 0$

→ How does log normal distribution mimic power law?

$$\ln \phi_Y(y) = \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \ln y - \frac{(\ln y - \mu)^2}{2\sigma^2}$$

If σ is very large such that

$|\ln y - \mu| << \sigma$

$$\ln \phi_Y(y) = \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \ln y$$

$$\boxed{\phi_Y(y) \sim \frac{1}{y} \text{ for large } y}$$

20/2/20. **Function of a R.V**

X is a r.v, Y, random

say $Y = \phi(X)$

(i) $\phi(x) = x^2$

(ii) $\phi(x) = \sqrt{x}$

(iii) $\phi(x) = \log x$

(iv) $\phi(x) = e^x$

pdf of X specified.

To find pdf of Y ?

Ex: Let X be a r.v with pmf

$P(X=-2) = \frac{1}{5}$

$P(X=-1) = \frac{1}{6}$

$P(X=0) = \frac{1}{5}$

$P(X=1) = 1/15$

$P(X=2) = 11/30$

Let $Y = X^2$

$A = \{-2,-1,0,1,2\}$

$B = \{0,1,4\}$

Now $P(Y=0) = \frac{1}{5}$

$P(Y=1) = \underbrace{\frac{1}{6} + \frac{1}{15}}_{P(X=1)+P(X=-1)} = \frac{7}{30}$

$P(Y=4) = P(X=2)+P(X=-2) = \frac{1}{5} + \frac{11}{30}$

$= \frac{17}{30}$

Ex: Let X be a Poisson with pmf

$$P[X=k] = \begin{cases} \frac{e^{-\lambda}\lambda^k}{k!}, & k=0,1,2,\ldots \\ 0 & \text{otherwise} \end{cases}$$

Let $Y = X^2+3$. Find $P(Y=y)$.

$\Rightarrow$ ~~Y maps~~ $A = \{0,1,2,\ldots\}$

onto $B = \{3,4,7,12,19,28,\ldots\}$

Inverse map $X = \sqrt{Y-3}$

only positive root bcoz X is taking only positive values

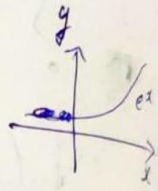
$P(Y=y) = P[X = \sqrt{Y-3}]$

$$P(Y=y) = \begin{cases} \frac{e^{-\lambda}\lambda^{\sqrt{y-3}}}{\sqrt{y-3}!}, & y \in B \\ 0 & \text{otherwise} \end{cases}$$

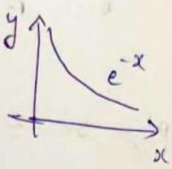
$\rightarrow$ ~~X is a rando~~

$Y = g(X)$

Eg: $Y = g(x) = e^x$, $x > 0$



Monotonic



$Y = g(x) = x^2$ for all x

$g'(x) = 2x \quad x > 0$

$= -2x \quad x < 0$

Here it is changing from +ve to -ve at $x = 0$.

$\Rightarrow$ Not monotonic.

If any function is monotonic, then below theorem can be used

<u>Theorem</u>: Let X be a r.v with pdf f. Let $y = g(x)$ be differentiable for all x and either $g'(x) > 0$ for all x (or) $g'(x) < 0$ for all x. Then $Y = g(X)$ is also a r.v with pdf $h[y]$.

modulus

$$h[y] = \begin{cases} f[g^{-1}(y)] \left|\frac{d}{dy} g^{-1}(y)\right| & , \alpha < y < \beta \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha = \min[g(-\infty), g(+\infty)]$

$\beta = \max[g(-\infty), g(+\infty)]$

<u>Proof</u>:

g is differentiable and $g'(x) > 0$ for all x, then g is continuous and strictly increasing.

The limits α and β exist (may be infinite) and inverse for $x = g^{-1}(y)$ exists.

cdf

$$P(Y \le y] = P[X \le g^{-1}(y)]$$

pdf of g is obtained on differentiation, we have

$$h(y) = \frac{d}{dy}(P(Y \le y)) = \frac{d}{dy} F_X[g^{-1}(y)]$$

$$= f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Similarly if $g' < 0$, then g is strictly decreasing

$\Rightarrow$ we have

$$P[Y \le y] = P[X \ge g^{-1}(y)]$$

$$= 1 - P[X \le g^{-1}(y)]$$

$$\cancel{h(y) = \frac{d}{dy}}$$

$$h(y) = -f_X[g^{-1}(y)] \cdot \frac{d}{dy}[g^{-1}(y)]$$

$\rightarrow$ <u>Ex</u>: $X \sim U(0,1)$

$$Y = -\frac{1}{\lambda} \ln(1-X)$$

$h(y) = ?$

Y is non-ve

$$f_X = \frac{1}{1-0} = 1$$

$$y = -\frac{1}{\lambda} \ln(1-x)$$

$$-\lambda y = \ln(1-x)$$

$$\Rightarrow 1 - x = e^{-\lambda y}$$

$$\Rightarrow 1 - e^{-\lambda y} = x$$

$$\Rightarrow g^{-1}(y) = 1 - e^{-\lambda y}$$

$$h(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

$$= 1 \cdot \left| \frac{d}{dy} (1 - e^{-\lambda y}) \right|$$

$$= 0 - e^{-\lambda y} \cdot (-\lambda)$$

$$h(y) = \lambda e^{-\lambda y}$$

$\rightarrow$ pdf

$\boxed{\text{(or)}}$

Y is non-ve

$F_Y(y) = 0 \quad , y \le 0$

$$F_Y(y) = P(Y \le y)$$

$$= P\left[\frac{-1}{\lambda} \ln(1-X) \le y \right]$$

$$= P[\ln(1-X) \ge -\lambda y]$$

$$= P[1 - X \ge e^{-\lambda y}]$$

$$= P[X \le 1 - e^{-\lambda y}]$$

$$\Rightarrow F_Y(y) = 1 - e^{-\lambda y}$$

$$f_Y(y) = \lambda e^{-\lambda y} \quad , 0 < y < \infty$$

$\downarrow$

pdf of Exp

$\rightarrow$ Let X have density

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(I) Let $Y = e^X$

$h(y) = ?$

(II) Let $Y = -2 \ln X$ $\quad h(y) = ?$

(I)

$$h(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

$$y = e^x$$

$$\Rightarrow \ln y = x$$

$$\Rightarrow g^{-1}(y) = \ln y$$

$$h(y) = 1 \cdot \frac{d}{dy} (\ln y)$$

$$h(y) = \begin{cases} \frac{1}{y} & 1 \le y \le e \\ 0 & \text{otherwise} \end{cases}$$

(II)

$$Y = -2 \ln X \qquad \left(\frac{-2}{x}\right)$$

$$x = e^{-Y/2}$$

$$h(y) = \left| \frac{-1}{2} \cdot e^{-Y/2} \right| \cdot 1$$

$$h(y) = \begin{cases} \frac{1}{2} e^{-Y/2} & 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

$\rightarrow$ Let X be a r.v

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \qquad -\infty < x < \infty$$

$$Y = X^2$$

$$h(Y) = ?$$

Here $y = g(x) = x^2$

$$g'(x) = 2x$$

$2x > 0 \ \forall \ x > 0$ & $2x < 0 \ \forall x < 0$

$\Rightarrow$ Theorem can't be applied

$Y = X^2$ For $y > 0$

$$F_Y(y) = P(Y \le y)$$
$$= P[X^2 \le y]$$
$$= P[-\sqrt{y} \le X \le y]$$
$$= P(X \le \sqrt{y}) - P(X \le -\sqrt{y})$$
$$= F_X(\sqrt{y}) - F_X(-\sqrt{y})$$
$$= F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$h(y) = \frac{1}{2\sqrt{y}}\left(f_X(\sqrt{y}) + f_X(-\sqrt{y})\right)$$
$$= \frac{1}{2\sqrt{y}}\left(\frac{1}{\sqrt{2\pi}} e^{-y/2} + \frac{1}{\sqrt{2\pi}} e^{-y/2}\right)$$

$$h(y) = \begin{cases} \dfrac{e^{-y/2}}{\sqrt{2\pi y}} & , y \ge 0 \\ 0 & , y < 0 \end{cases}$$

cf χ_1^2 chisquare distribution.